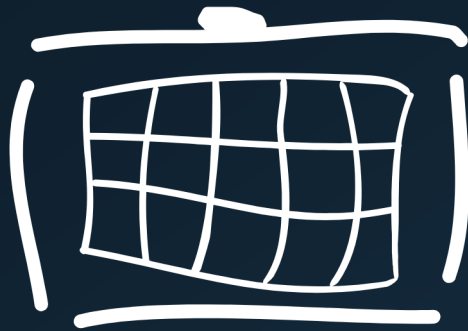




Radiator

CONFIGURATION GUIDE

Okta Application Setup

Date: 2026-08-26

This configuration allows authenticating RADIUS users from Radiator Policy Server against Okta using the OAuth2 Resource Owner Password Credentials (ROPC) flow. Radiator exchanges the user's credentials for an access token via Okta's Authorization server, then queries the `/userinfo` endpoint to retrieve the user's profile and group memberships.

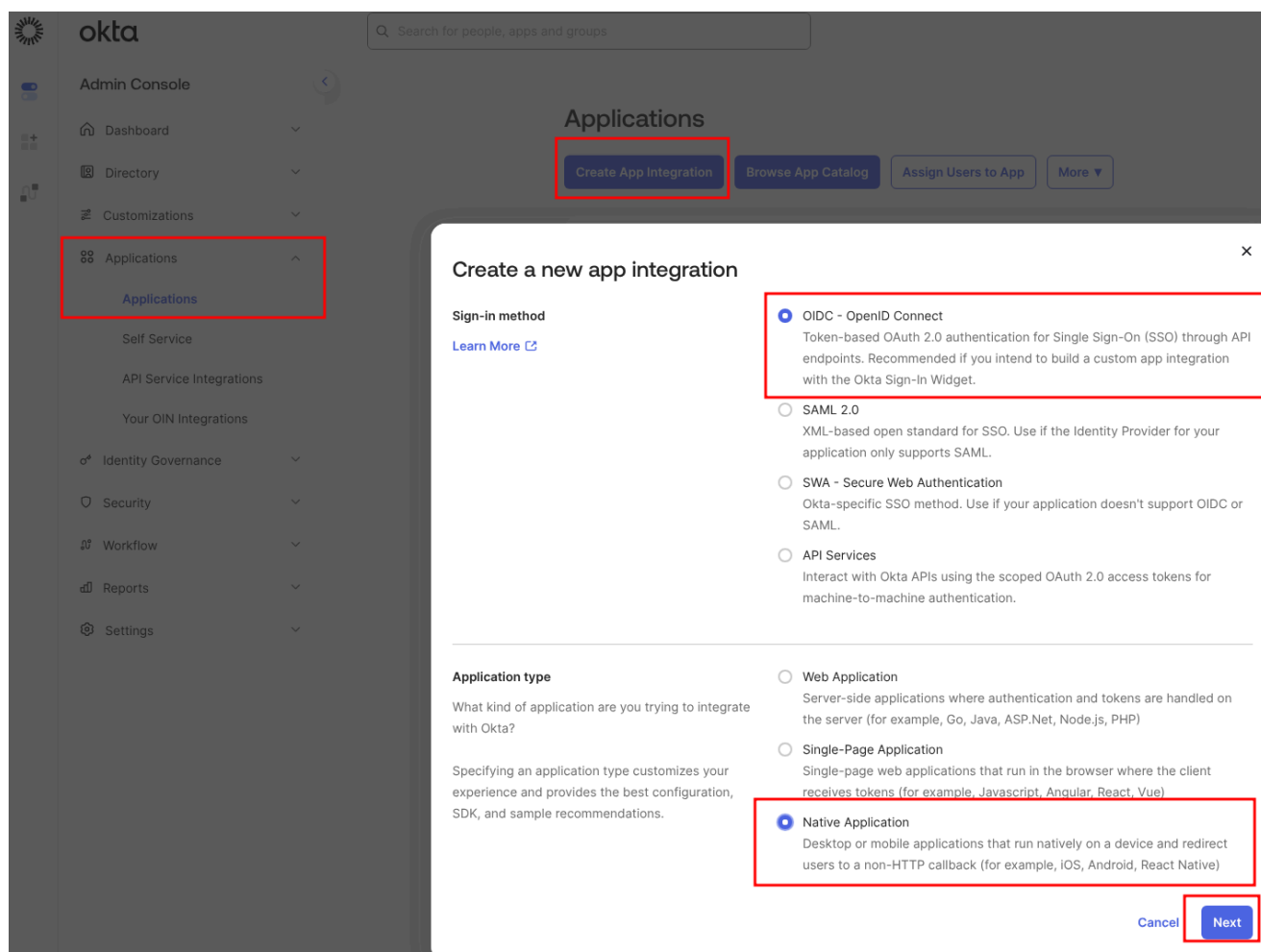
NOTE Using ROPC and adding a custom claim to Authorization server requires the Okta API Access Management add-on.

Prerequisites

- An Okta tenant
- User accounts, optionally belonging to groups
- Admin access to the Okta Admin Console

Create an OIDC Application

1. Go to **Applications > Applications > Create App Integration**
2. Select **OIDC - OpenID Connect**
3. Select **Native Application** as application type



4. Configure the application:
 - App integration name: e.g., Radiator ROPC
 - Grant type: Check **Resource Owner Password** (under "Advanced" > "Other grants")
 - Sign-in redirect URIs: `http://localhost` (required, but not used by Radiator)
 - Assignments: Either **Allow everyone in your organization to access** or **Limit access to selected groups** depending on your needs

5. Click **Save**

New Native App Integration

General Settings

App integration name

Radiator ROPC

Proof of possession

☐ Require Demonstrating Proof of Possession (DPoP) header in token requests

Grant type

Core grants

☒ Authorization Code

☐ Refresh Token

☐ Device Authorization

Advanced >

These grants are more sensitive and should be enabled only if necessary.

Non-interactive grants

☐ SAML 2.0 Assertion

☐ Token Exchange

Okta direct auth API grants

☐ OTP

☐ OOB

☐ MFA OTP

☐ MFA OOB

Other grants

☒ Resource Owner Password

☐ Implicit (hybrid)

Sign-in redirect URIs

☐ Allow wildcard * in sign-in URI redirect.

Okta sends the authentication response and ID token for the user's sign-in request to these URIs

http://localhost

+ Add URI

Learn More

Sign-out redirect URIs (Optional)

+ Add URI

Learn More

Assignments

Controlled access

☒ Allow everyone in your organization to access

☐ Limit access to selected groups

☐ Skip group assignment for now

Select whether to assign the app integration to everyone in your org, only selected group(s), or skip assignment until after app creation.

Enable immediate access (Recommended)

☒ Enable immediate access with Federation Broker Mode

To ensure optimal app performance at scale, Okta End User Dashboard and provisioning features are disabled. Learn more about Federation Broker Mode.

Recommended if you want to grant access to everyone without pre-assigning your app to users and use Okta only for authentication.

Save

Cancel

Configuration guide


Radiator

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6. Create **Client secret** from the application's **General** tab

- Select **Edit**

The screenshot shows the 'Radiator ROPC' application configuration page in the Okta Admin Console. The 'General' tab is selected. In the 'Client Credentials' section, the 'Edit' button is highlighted with a red box. The 'Client ID' is 'Ooa10uym220oEvt0P698'. The 'Client authentication' options are 'None' (selected), 'Client secret', and 'Public key / Private key'. The 'Proof Key for Code Exchange (PKCE)' option is 'Require PKCE as additional verification' (checked).

- Select Client secret (under client authentication)
- Click **Save** to have client secret created

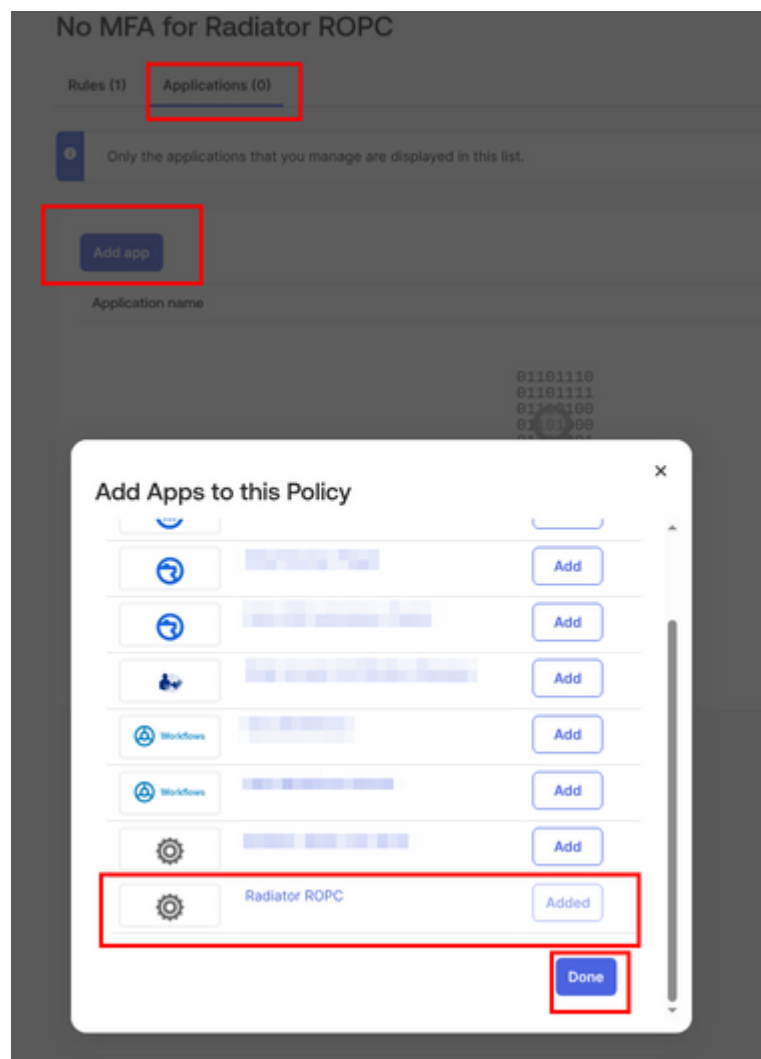
The screenshot shows the 'Radiator ROPC' application configuration page in the Okta Admin Console. The 'General' tab is selected. In the 'Client Credentials' section, the 'Client secret' option is selected under 'Client authentication' and is highlighted with a red box. The 'Save' button at the bottom is also highlighted with a red box. The 'Generate new secret' button is visible above the 'CLIENT SECRETS' table.

7. Note **Client ID** and **Client secret** down, these are needed on the Radiator configuration side (see [Radiator Okta configuration guide](#))

Disable MFA

The ROPC flow does not support MFA. If MFA is enabled, the token request will fail. Ensure MFA is not enforced for the users or groups:

1. Go to **Security > Authenticators**
2. Review active authenticators: if any MFA factors (e.g., Okta Verify, SMS) are set to **Required**, the ROPC flow will be blocked
3. Go to **Security > Authentication Policies** (or **Security > Authentication > App Sign-in Policies** in classic UI)
4. Create a new policy for the ROPC app:
 - Select **Create Policy** and name the new policy e.g. `No MFA for Radiator ROPC`
 - Add Radiator ROPC application to the policy from **Applications** tab. **Add app**, select your app and select **Done**



- Add a rule that removes the requirement for MFA
- Select **Add rule** and set **AND Possession factor constraints are to Not required**
- Alternatively: set the authentication requirements in **AND User must authenticate with to Password only (1FA)**

Rule name

No MFA

IF

IF

User's user type is

Any user type

AND

User's group membership includes

Any group

AND

User is

Any user

AND

Device state is

☒ Any

☐ Registered

 Setup Okta Verify as [Authenticator](#)

AND

Device assurance policy is

No policy

AND

Device platform is

Any platform

Device platform is securely verified for registered devices. For unregistered devices, it's determined by the user-agent and can be modified. Learn more about [Device platform security](#)

AND

User's IP is

Any IP

AND

Risk is

Any

AND

The following custom expression is true

This is an optional advanced setting. If the expression is formatted incorrectly or conflicts with conditions set above, the rule may not match any users.

[Expression language reference](#)

THEN

THEN

Access is

☐ Denied

☒ Allowed after successful authentication

AND

User must authenticate with

Any 2 factor types

AND

Possession factor constraints are

☒ Phishing resistant

☐ Hardware protected

☐ Require user interaction

☒ Any interaction

 This includes responding to an approval prompt in Okta Verify, or touching an Yubikey to activate.

☐ Require device passcode or biometric user verification

 This can be met by any operating system and achieves 2FA with a single authenticator

☐ Require biometric user verification

 This can only be met by Okta Verify FastPass or Okta Verify Push on Android, iOS, and macOS.

Learn more about [possession factor constraints](#)

AND

Authentication methods

☒ Allow any method that can be used to meet the requirement

☐ Disallow specific authentication methods

☐ Allow specific authentication methods

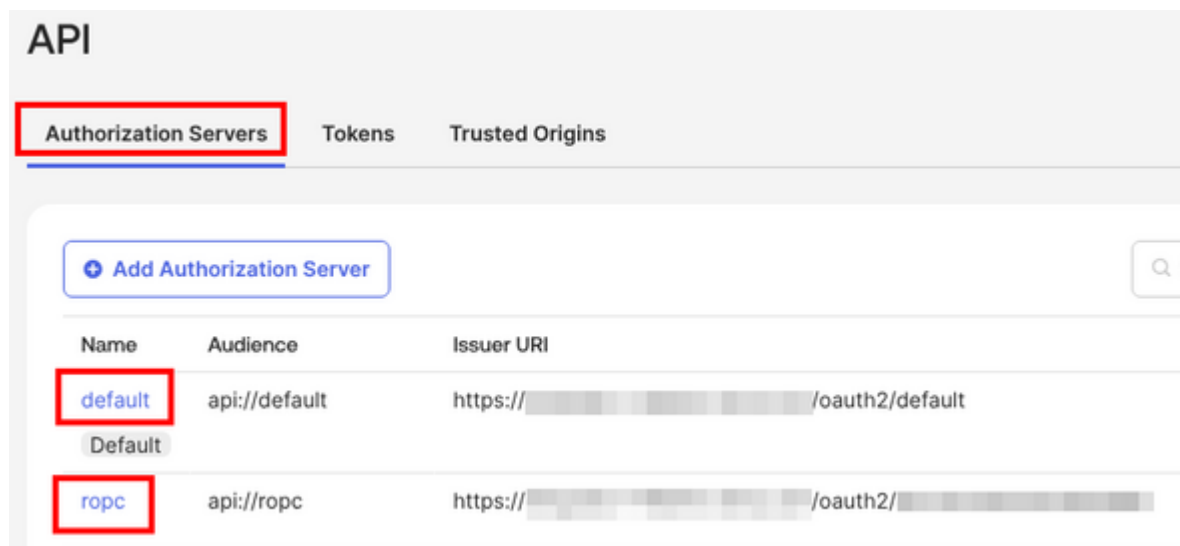
Configure the Authorization Server

The ROPC flow uses Okta's Authorization server. This can be either the Default Authorization server (/oauth2/default) or custom Authorization Server. The groups claim must be added to the used Authorization Server so that group memberships appear in the /userinfo response.

NOTE Using ROPC and adding a custom claim to Authorization server requires the Okta API Access Management add-on.

- Go to **Security > API > Authorization Servers**

2. Click on **default** or your custom Authorization Server



3. Go to the **Claims** tab

4. Click **Add Claim**:

- Name: `groups`
- Include in token type: **Access Token**
- Value type: **Groups**
- Filter: **Matches regex** `.*` (this means include all groups, more restrictive filter is possible)
- Include in: **Any scope**

5. Click **Create**

The screenshot shows the 'Add Claim' form for the 'default' authorization server. The 'Claims' tab is selected. The form fields are as follows:

- Name: `groups`
- Include in token type: **Access To...** (dropdown), **Always** (dropdown)
- Value type: **Groups** (dropdown)
- Filter: **Matches regex** (dropdown), `.*` (text input)
- Disable claim: ☐ Disable claim
- Include in: ☒ Any scope, ☐ The following scopes:

Buttons: **Create**, **Cancel**

6. Repeat steps 4 and 5 to token type **ID Token**

default

Active ▾

Default

Settings

Scopes

Claims

Access Policies

Token Preview

+ Add Claim

Claim type	Name	Value	Scopes	Type	Included	
All	sub	(appuser != null) ? appuser.userName : app.clientId	Any	access	Always	
ID	groups	groups: matches regex .*	Any	access	Always	
Access	groups	groups: matches regex .*	Any	id	Always	

Add Access Policy for ROPC

The Authorization Server needs an access policy that allows the ROPC grant type for your application:

1. Go to **Security > API > Authorization Servers**
2. Click on **default** or your custom Authorization Server
3. Go to the **Access Policies** tab

4. Either edit the existing policy or create a new policy:
- If editing: assign it to the **Radiator ROPC** application
 - If creating: give it a name (e.g., `ROPC Policy`) and assign it to the **Radiator ROPC** application

The screenshot shows the 'default' configuration page with tabs for Settings, Scopes, Claims, Access Policies, and Token Preview. The 'Access Policies' tab is selected and highlighted with a red box. Below the tabs is a button labeled 'Add New Access Policy'. An 'Edit Policy' modal is open, showing fields for Name, Description, and Assign to. The Name field contains 'ROPC policy', the Description field contains 'Policy for ROPC client', and the Assign to field has radio buttons for 'All clients' and 'The following clients:'. The 'The following clients:' option is selected, and a list box below it contains 'Radiator ROPC' with a close button (x). At the bottom of the modal are 'Update Policy' and 'Cancel' buttons.

default

Active ▾ Default

Settings Scopes Claims **Access Policies** Token Preview

Add New Access Policy

Edit Policy

Name

Description

Assign to

☐ All clients

☒ The following clients:

Update Policy Cancel

5. Add or edit a rule within the policy:
- Grant type: ensure **Resource Owner Password** is checked (under Advanced)

Edit Rule

Rule Name
ROPC

IF Grant type is

Client acting on behalf of itself

- ☒ Client Credentials

Core grants

- ☒ Authorization Code
- ☒ Device Authorization
- Advanced >**

Non-interactive grants

- ☐ SAML 2.0 Assertion
- ☐ Token Exchange

Okta direct auth API grants

- ☐ OTP
- ☐ OOB
- ☐ MFA OTP
- ☐ MFA OOB

Other grants

- ☐ Implicit (hybrid)
- ☒ **Resource Owner Password**
- ☐ Client-initiated backchannel authentication flow (CIBA)

AND User is

- ☒ Any user assigned the app
- ☐ Assigned the app and a member of one of the following:

AND Scopes requested

- ☒ Any scopes
- ☐ The following scopes:

THEN Use this inline hook

None (disabled)

6. Click **Create Rule / Update Rule**

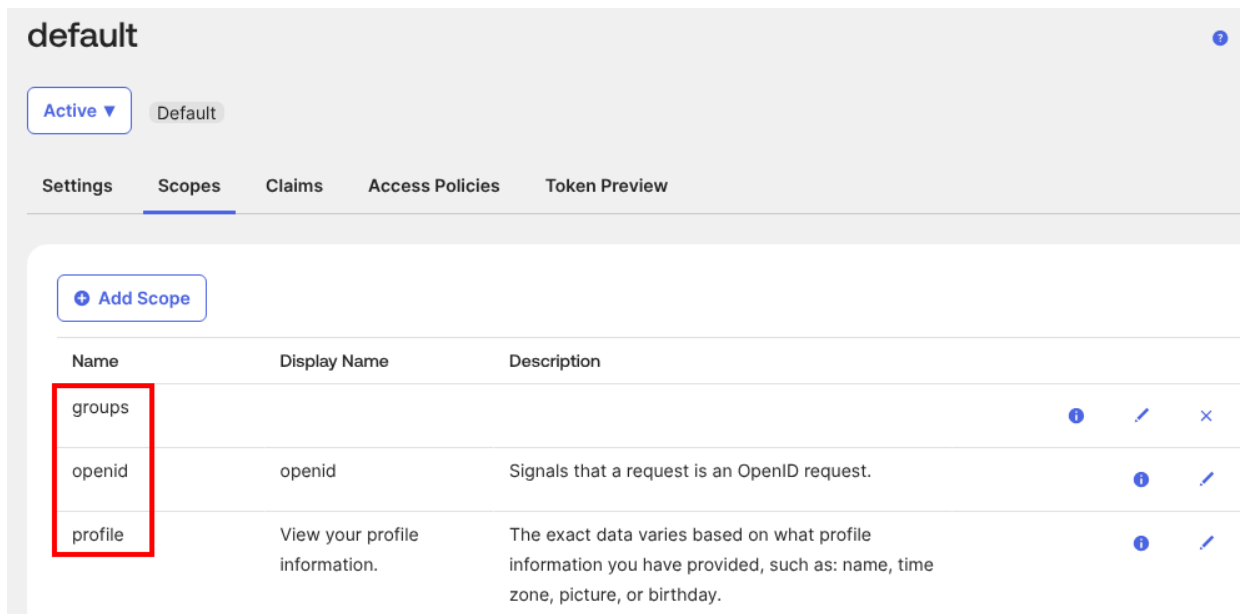
Without this policy, token requests will fail even if the application itself has the ROPC grant type enabled.

Verify Scopes

1. In the Authorization Server, go to the **Scopes** tab

2. Ensure these scopes exist:

- openid
- profile
- groups Add any scope that does not exist: click **Add Scope**, name it e.g. groups, select **Create**



Radiator Policy Server configuration

After completing the previous steps you will need to update your Radiator Policy Server configuration. See [Radiator Policy Server Okta configuration guide](#) for more details.

Troubleshooting

Here is a collection of known errors shown on the Radiator Policy Server authentication log, and how they can be fixed either on Radiator Policy Server configuration or on Okta side. When Radiator Policy Server configuration is edited, remember to restart Radiator Policy Server after changes

Error: The credentials provided were invalid

```
{"error": "invalid_grant", "error_description": "The credentials provided were invalid."}
```

Solution

Check both password and username as either one or both are incorrect

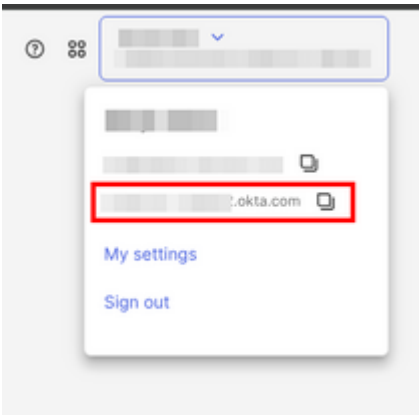
Error: Invalid value for 'client_id' parameter

```
{"errorCode": "invalid_client", "errorSummary": "Invalid value for 'client_id' parameter."}
```

Solution

There are multiple solutions to this error:

- Check that Radiator Policy Server configuration and the `vars.aaa.oauth2.issuer-uri` and the `vars.aaa.oauth2.authz-server` have the correct Okta domain and Authorization Server identifier



API

Authorization Servers Tokens Trusted Origins

+ Add Authorization Server

Search

Name	Audience	Issuer URI
default	api://default	https://[redacted]okta.com/oauth2/[redacted]default
Default		
ropc	api://ropc	https://[redacted].okta.com/oauth2/[redacted]

- Check that Radiator Policy Server configuration and the `vars.aaa.oauth2.client` parameter has the correct Client ID

← Back to Applications

Radiator ROPC

Active ▾ View Logs

General Sign On Assignments Okta API Scopes Application Rate Limits

Client Credentials Edit

Client ID 00a10uym220oEvt0P698 📋

Public identifier for the client that is required for all OAuth flows.

Client authentication

☐ None

☒ Client secret

☐ Public key / Private key

Proof Key for Code Exchange (PKCE) ☒ Require PKCE as additional verification

CLIENT SECRETS

Generate new secret

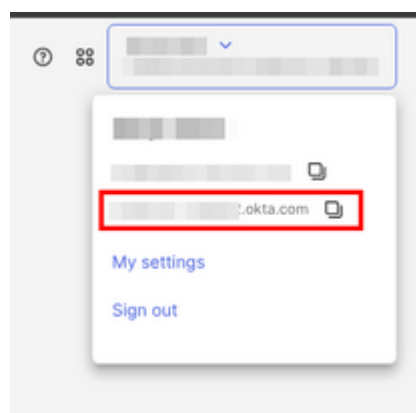
Creation date	Secret	Status
Mar 10, 2026	 👁	📋 Active ▾

Error: 401 Unauthorized

"Reason": "HTTP backend 'OKTA_ROPC' query 'AUTH_USERS' failed with status '401 Unauthorized':"

Solution

Check that Radiator Policy Server configuration and the `vars.aaa.oauth2.okta-domain` has the correct Okta domain



Error: invalid_client

```
{"error": "invalid_client", "error_description": "The client secret supplied for a confidential client is invalid."}
```

Solution

Check the Radiator Policy Server configuration and the `vars.aaa.oauth2.secret` parameter has the secret value from the correct application.

← Back to Applications



Radiator ROPC

Active ▾

View Logs

General

Sign On

Assignments

Okta API Scopes

Application Rate Limits

Client Credentials

Edit

Client ID

00a10uym220oEvt0P698



Public identifier for the client that is required for all OAuth flows.

Client authentication

☐ None

☒ Client secret

☐ Public key / Private key

Proof Key for Code Exchange (PKCE)

☒ Require PKCE as additional verification

CLIENT SECRETS

Generate new secret

Creation date	Secret	Status
Mar 10, 2026	 	Active ▾ 

Error: 400 Bad Request

"Reason": "HTTP backend 'OKTA_ROPC' query 'AUTH_USERS' failed with status '400 Bad Request'"

Solution

Check that Radiator Policy Server configuration and the `vars.aaa.oauth2.authz-server` has the correct Authorization Server identifier

API

Authorization Servers Tokens Trusted Origins

+ Add Authorization Server

Search

Name	Audience	Issuer URI
default	api://default	https://[redacted]okta.com/oauth2/default
Default		
ropc	api://ropc	https://[redacted].okta.com/oauth2/[redacted]